

5 Bit Successive Approximation Register ADC

EE23BTECH11027¹ and EE23BTECH11042²

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Abstract

The lab report detailed the design, implementation, and analysis of a 5-bit Successive Approximation Register Analog-to-Digital Converter circuit. The SAR ADC works with some iteration steps to approximate an analog voltage by estimating the highest bit first and moving down to the lowest until some desired degree of accuracy for a 5-bit digital output is reached. Three stages were key in this design: the DAC used to generate the digital representation of the desired analog voltage, the comparator to compare the two voltages, and the SAR logic to generate the control signals for this process.

1. Introduction

The design and implementation of a 5-bit Successive Approximation Register (SAR) Analog-to-Digital Converter (ADC) are described. The SAR ADC uses SAR logic realized by D flip-flops, an R-2R ladder network for the DAC, and a comparator to perform the analog-to-digital conversion. A sampling circuit is also included. The building blocks are explained in the following sections:

For beginning the ADC cycle, let the most significant bit of the digital output be 1, and all other bits are 0. The SAR logic comes to a decision regarding the MSB digital output, based on the conclusion of the comparison in the comparator. If the output of the comparison circuit is 1, the MSB digital output is not altered. When the output of the comparison circuit is 0, the MSB digital output does become 0. The clocks needed to complete a 5-bit SAR ADC.

4. DAC

The DAC in our SAR ADC is an R-2R ladder network, as shown in Figure 3, which uses resistors with values of R and $2R$ to create appropriate reference voltages to get the proper output. Each digital bit D_n from the SAR logic feeds into the R-2R network, which scales the output voltage proportionally to the binary code. The DAC's output is connected to the comparator for comparison with the analog input.

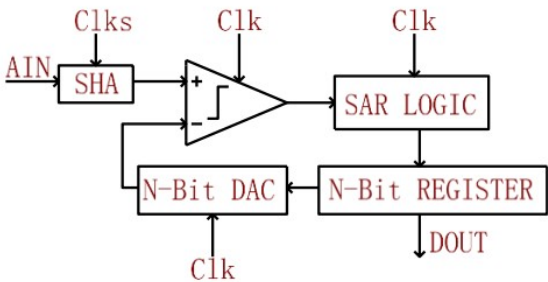


Figure 1. SAR LOGIC

2. Components Used

Component	Remarks	Quantity
IC7474	D Flip-flop	6
Resistors	1kΩ and 2kΩ	4, 6
LM358	Dual Op-amp	2
Microcontroller	Arduino Uno	1
Capacitor	104 pF	1

3. SAR LOGIC

Figure 3 illustrates the following circuit. The circuit consists of a flip-flop. The EN signal is high while the clock cycles to 0. In this instance, all of the outputs of the flip-flop are reset to 0, and EN is low during the other clock cycles. The MSB determines the trigger, according to the cyclical clock action, for the digital output of D4, D3, D2, D1, D0 based on the output of the comparator.

To minimize the number of registers, an unrreredundant SAR logic is proposed. For a given N-bit SAR AD Converter, N registers are needed. The register acts as a shift, input data, and storage.

The output voltage of the DAC can be calculated as:

$$V_{out} = V_{ref} \times \frac{D}{2^N}$$

where D is the digital code, N is the bit resolution (5 in this case), and V_{ref} is the reference voltage. The resistor values ensure that each successive bit contributes half of the previous bit's value to the output.

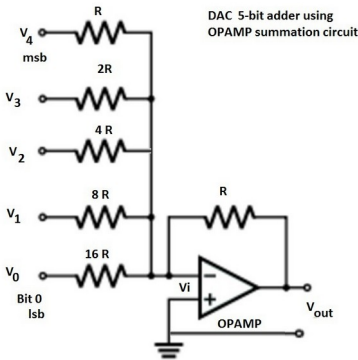


Figure 2. DAC R-2R ladder

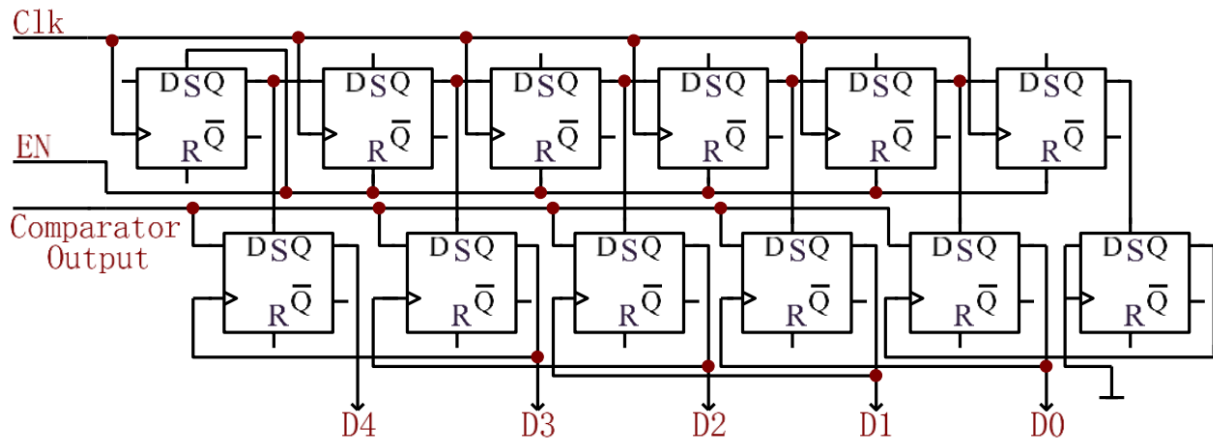


Figure 3. SAR Logic

5. Calculations for DAC

For an R-2R ladder with a reference voltage V_{ref} , the output V_{out} for a digital code $D = D_4D_3D_2D_1D_0$ is given by the weighted sum:

$$V_{out} = V_{ref} \times \left(\frac{D_4}{2} + \frac{D_3}{4} + \frac{D_2}{8} + \frac{D_1}{16} + \frac{D_0}{32} \right)$$

Using 1 k Ω and 2 k Ω resistors provides accurate division of V_{ref} , allowing each bit to influence V_{out} precisely. This design ensures that the DAC meets the 5-bit resolution, outputting accurate voltages each binary input combination. (Here, D_4 is MSB while D_0 is LSB)

6. Working

We have used Arduino to give the clock (clk) and the enable signal EN, as mentioned in the figure 3. The IC7474 has NOT gate for both the Preset and set (Here, S and R respectively). Thus, instead of connecting the Presets (S) of the bottom flip-flops to Q, we connected

7. Result and Conclusion

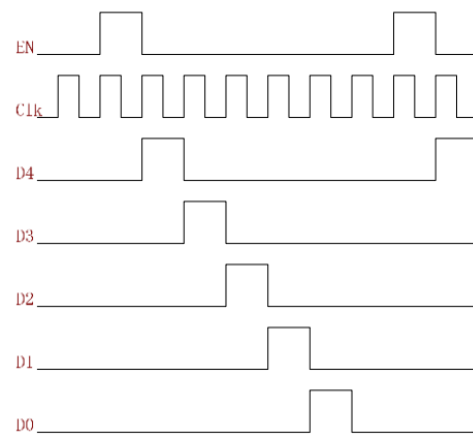


Figure 4. SAR Logic timing diagram

The SAR logic is the control unit that performs the binary search algorithm. It initializes the MSB and compares the DAC output as whether it is greater or lesser than the analog input using the comparator's output. If the DAC output is less than the analog input, then the bit remains set; otherwise, it gets cleared. The SAR logic now advances to the next bit in sequence and follows similar steps for each bit, until it has obtained its approximation. The whole process then continues, iteratively from the MSB to the LSB (Least Significant Bit), to derive the binary representation of the analog input.

The N-bit register contains the result of the SAR logic as the final digital output D_{OUT} of the ADC. After all the bits are decided by the successive approximation process, the N-bit register has retained the value that best approximates the given analog input. The D_{OUT} signal is registered to the clock Clk ; it triggers the reading of the digital output once the conversion is done.

As the timing diagram reveals, the implemented 5-bit SAR ADC operates based on the proposed scheme. One bit at a time, all the bits (D_4 to D_0) are set or cleared by the clock and enable signals: both appear as part of the sampling and conversion phases. The sequence and stability of bits depict the SAR logic binary search algorithm where each bit decision is saved for subsequent comparisons from the most significant bit (MSB) downwards to the least significant bit (LSB). This is an approximate bit from D_4 to D_0 that illustrates the fact that SAR ADC produces a precise digital output which correctly represents the input. Hence, the performance observed by the timing diagram confirms that SAR ADC does have effective performance since it is possible to result in a reliable and accurate representation of the input analog signal as a digital one.

■ References

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